UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Origin of Electrode Potential and Cell Potential

Complete study pack for class 001: galvanic and electrolytic cells, electrode potential and the Helmholtz double layer, standard electrode potential, cell notation, cell potential, $\Delta G = -nFE$ and the electrochemical series

Compiled from the PESU Unit 2 class 001 slide deck and the first 19 minutes of its AV-summary transcript, cross-checked against the consolidated Unit 2 slides, with past ESA papers for weighting and problems. Generated 17 October 2026.

· How to use this pack

Topics 1–5 follow the lecture. The Nernst equation, taught in the second half of the same video, is the subject of the class 002 pack. Cell numericals with non-standard concentrations (Nernst) are in classes 002 and 006; here every worked example stops at E°_{cell} and ΔG°. Learn F4.1 and W4.1 first: the same steps open every ESA cell problem. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·10	Deck 001, slide 10 (rendered .pptx)
T001 00:08:22	AV-summary transcript 001, segment start
ESA Jan-25 s2 2b	Past paper: Jan 2025 set 2, question 2.b
ESA Jul-23a 4d	Past paper: Jul 2023 (first UE22 paper), question 4.d

Contents

1. 1 · Electrochemistry and the two types of cell
2. 2 · Electrode potential and its origin: the Helmholtz double layer
3. 3 · Standard electrode potential, electrochemical cell and cell notation
4. 4 · Cell potential, standard cell potential and free energy
5. 5 · The electrochemical series and its uses
6. Appendix A · Symbol glossary
7. Appendix B · Sources, coverage and gaps

1 Electrochemistry and the two types of cell

Electrochemistry deals with the interconversion of chemical energy and electrical energy. A galvanic cell converts the chemical energy of a spontaneous redox reaction into electrical energy (batteries, fuel cells); an electrolytic cell uses electrical energy to drive a non-spontaneous redox reaction (electroplating, electrorefining). Every electrochemical system needs three things: a redox reaction, electrodes (anode for oxidation, cathode for reduction) and an electrolyte that conducts by the movement of ions.

Key terms and definitions

Electrochemistry. The branch of chemistry that deals with the interconversion of chemical energy and electrical energy. S001·3 · T001 00:00:45

Galvanic (voltaic) cell. A cell that converts chemical energy to electrical energy, using a spontaneous redox reaction; e.g. batteries and fuel cells. S001·3 · T001 00:01:15

Electrolytic cell. A cell that converts electrical energy to chemical energy: an external supply drives a non-spontaneous redox reaction; e.g. cells used in electroplating and electrorefining. S001·3 · T001 00:02:19

Anode / cathode. The anode is the electrode where oxidation takes place; the cathode is the electrode where reduction takes place. S001·4 · T001 00:03:30

Electrolytic conductance. Conduction through the electrolyte by the movement of ions (electrons cannot move through the electrolyte). Electrolytes may be acid, alkali or salt solutions, molten electrolytes or solid electrolytes. S001·4 · T001 00:03:49

Graphs and diagrams

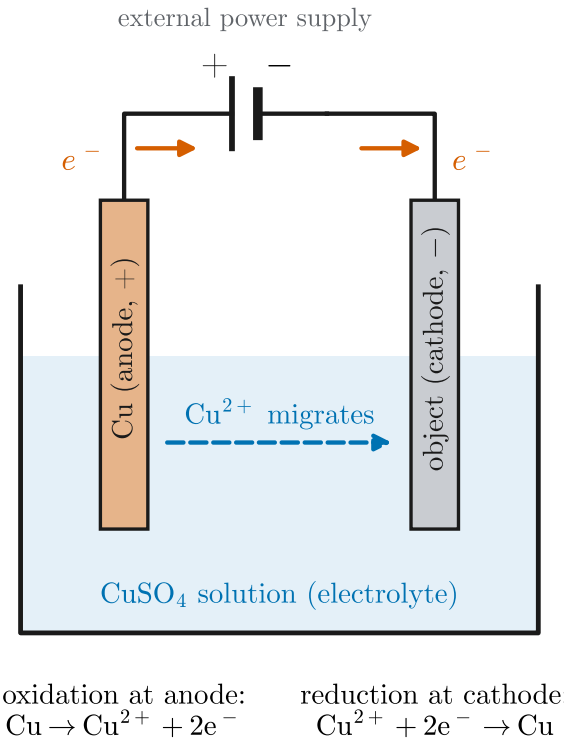
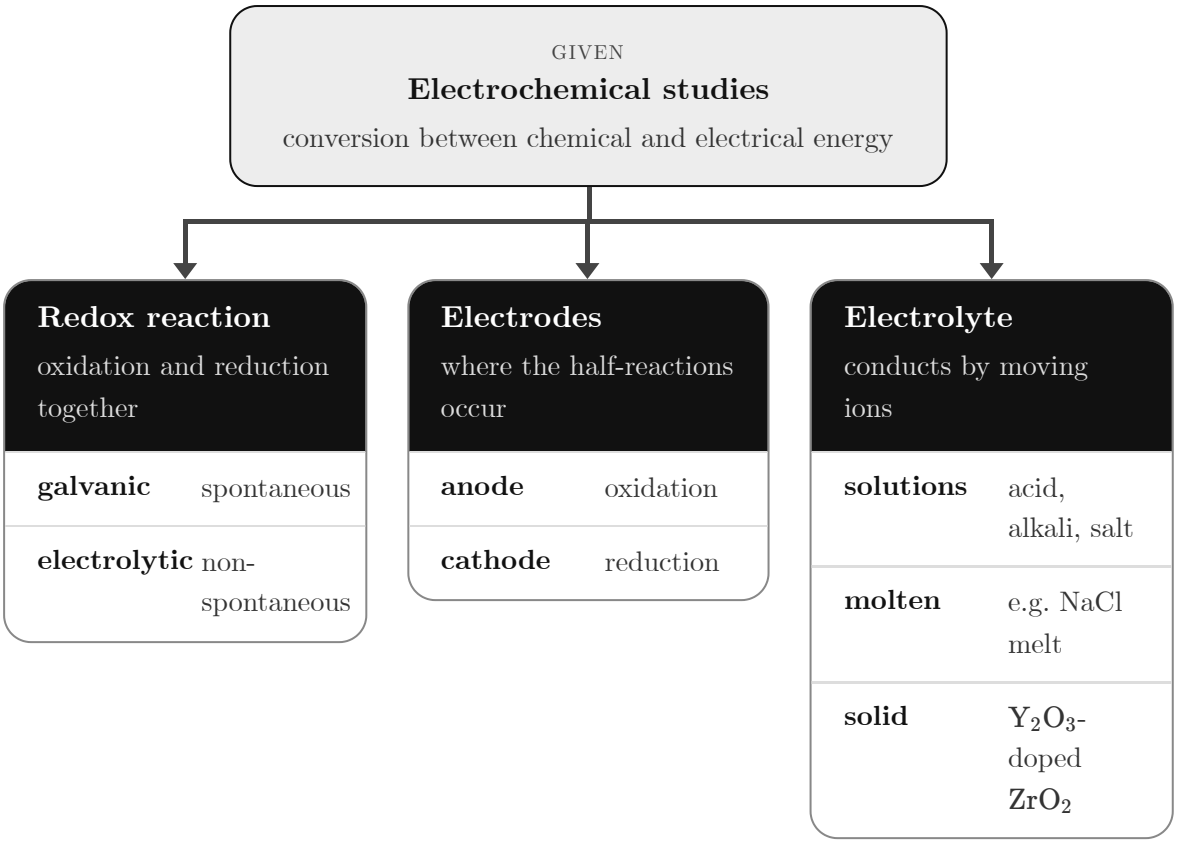


Figure G1.1. An electrolytic cell (copper electroplating). ESA ★

ADDED FOR CLARITY S001·3 · T001 00:02:50

How to read it. The external supply pushes electrons into the object (cathode, −), where Cu^{2+} is reduced and plated; the copper anode (+) is oxidised and replaces the ions. Compare with the galvanic Daniell cell (G3.1): there the reaction itself drives the electrons and the anode is negative.

Flowcharts



Remark. The molten and solid examples come from the lecture (T001 00:04:28–00:05:32).

Figure F1.1. What every electrochemical system needs ESA ★

S001·4 · T001 00:02:58

Galvanic cell versus electrolytic cell

	Galvanic cell	Electrolytic cell
Energy conversion	chemical → electrical	electrical → chemical
Redox reaction	spontaneous	non-spontaneous (driven)
ΔG, E_{cell}	$\Delta G < 0$, $E_{cell} > 0$	$\Delta G > 0$, $E_{cell} < 0$ for the driven reaction
Anode	oxidation; negative terminal	oxidation; connected to + of the supply
Cathode	reduction; positive terminal	reduction; connected to − of the supply
Salt bridge	usually two solutions joined by a salt bridge	usually one vessel, no salt bridge
Examples	batteries, fuel cells, Daniell cell	electroplating, electrorefining, metallurgy

ADDED FOR CLARITY S001·3 · T001 00:01:15 · T001 00:02:19

Important points and why

1. A galvanic cell needs a **spontaneous** redox reaction, because no energy is supplied from outside; an electrolytic cell drives a **non-spontaneous** one with external electrical energy. ESA ★★ T001 00:01:30 · T001 00:02:34
2. Oxidation and reduction always happen simultaneously; producing electrical energy needs an exchange of electrons, hence a redox reaction. ESA ★ S001·4 · T001 00:03:16
3. Electrons cannot move through the electrolyte; only ions carry the current there. ESA ★★ T001 00:03:57
4. Solid electrolyte example: zirconia doped with Y_2O_3 has oxide-ion vacancies, so O^{2-} ions move through the solid. ESA ★ TRANSCRIPT ONLY T001 00:04:58
5. Molten electrolytes work because melting (e.g. NaCl) frees the ions to move. ESA ★ TRANSCRIPT ONLY T001 00:04:28

SHORT-NOTE RECAP

Electrochemistry: interconversion of chemical and electrical energy. Galvanic cell: spontaneous redox reaction → electrical energy (batteries, fuel cells). Electrolytic cell: external energy drives a non-spontaneous reaction (electroplating). Needs: a redox reaction; anode (oxidation) and cathode (reduction); an ionic electrolyte (solution, melt or solid).

2 Electrode potential and its origin: the Helmholtz double layer

When a metal rod is dipped in a solution of its own ions, a potential develops at the metal–solution interface: the electrode potential E. It arises because two opposite reactions compete, ionisation ($\mathrm{M} \rightarrow \mathrm{M^{n+}} + \mathrm{ne^-}$) and deposition ($\mathrm{M^{n+}} + \mathrm{ne^-} \rightarrow \mathrm{M}$). Whichever is faster charges the metal (negative in case I, positive in case II); the charged metal attracts a layer of oppositely charged ions, forming the Helmholtz electrical double layer, and an equilibrium is set up.

Key terms and definitions

Electrode potential (E). When a metal rod is dipped in a solution of its own ions, the electrical potential developed at the interface of the metal and its solution. It is denoted by E. S001·4 · T001 00:05:32

Helmholtz electrical double layer. The layer of charge on the metal surface together with the layer of oppositely charged ions it attracts from the solution, formed at the metal–solution interface; it is responsible for the electrode potential. S001·6 · S001·7 · T001 00:08:22

> *"When a metal M is in contact with solution containing its ions* $\mathrm{M^{n+}}$*, two reactions are possible: 1. Ionisation (oxidation)* $\mathrm{M} \rightleftharpoons \mathrm{M^{n+}} + \mathrm{ne^-}$*; 2. Deposition (reduction)* $\mathrm{M^{n+}} + \mathrm{ne^-} \rightleftharpoons \mathrm{M}$*."* S001·5

> *"Case I: if ionization is faster than deposition, the metal acquires net negative charge, consequently retards the rate of ionization and increases the rate of deposition. This ultimately leads to the establishment of equilibrium. The metal electrode gets negatively charged and attracts the layer of positive ions at the interface; an electrical double layer is formed at the interface of metal and solution known as Helmholtz electrical double layer."* S001·6

> *"Case II: if deposition is faster than ionization, the metal acquires net positive charge, consequently retards the rate of deposition and increases the rate of ionization. This ultimately leads to the establishment of equilibrium. The metal electrode gets positively charged and attracts the layer of negative ions at the interface; an electrical double layer is formed, known as Helmholtz electrical double layer."* S001·7

Formulas

E2.1 · The two competing electrode reactions ESA ★★

$$\underbrace{\mathrm{M} \rightleftharpoons \mathrm{M^{n+}} + n\mathrm{e^-}}_{\text{ionisation (oxidation)}} \qquad \underbrace{\mathrm{M^{n+}} + n\mathrm{e^-} \rightleftharpoons \mathrm{M}}_{\text{deposition (reduction)}}$$

GIVES	The origin of the electrode potential: the faster reaction decides the sign of the charge on the metal.
APPLIES	A metal M in contact with a solution of its own ions.
UNITS	—
NOTE	Ionisation faster: metal negative (case I). Deposition faster: metal positive (case II).

S001·5 · S001·6 · S001·7

Graphs and diagrams

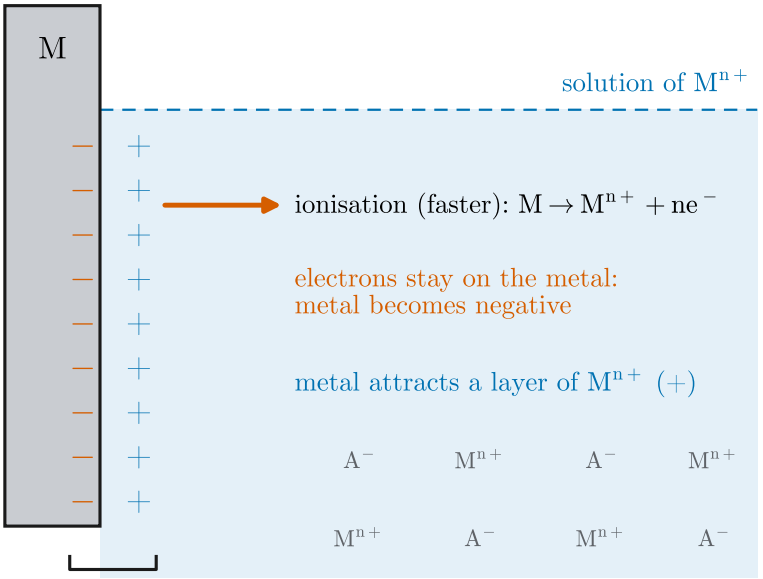


Figure G2.1. Case I: ionisation faster than deposition. ESA ★★★

S001·6 · T001 00:07:29 · ESA Jan-25 s2 2b

How to read it. Redrawn from slide 001·6. Draw the metal rod, a column of − charges on its surface, a parallel column of + ions in the solution, and label the pair "Helmholtz electrical double layer". Write $\mathrm{M} \rightleftharpoons \mathrm{M^{n+}} + \mathrm{ne^-}$ beside it.

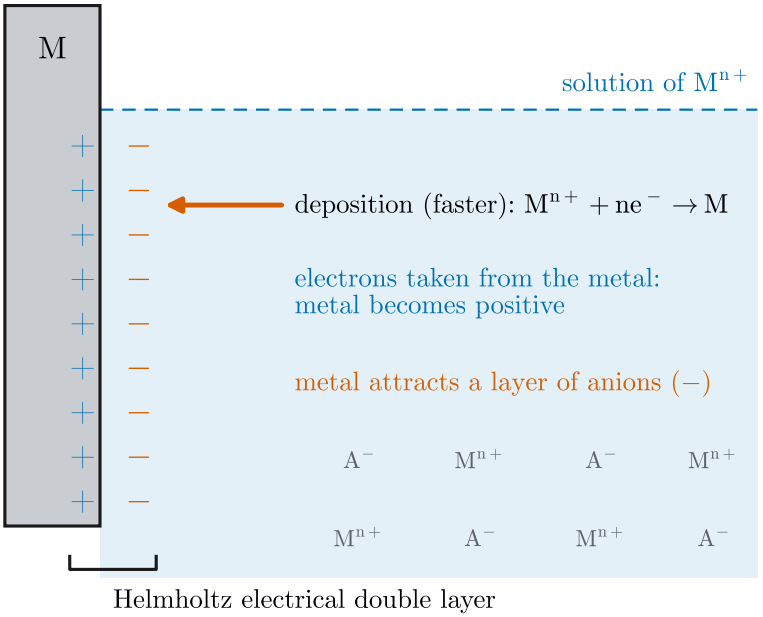
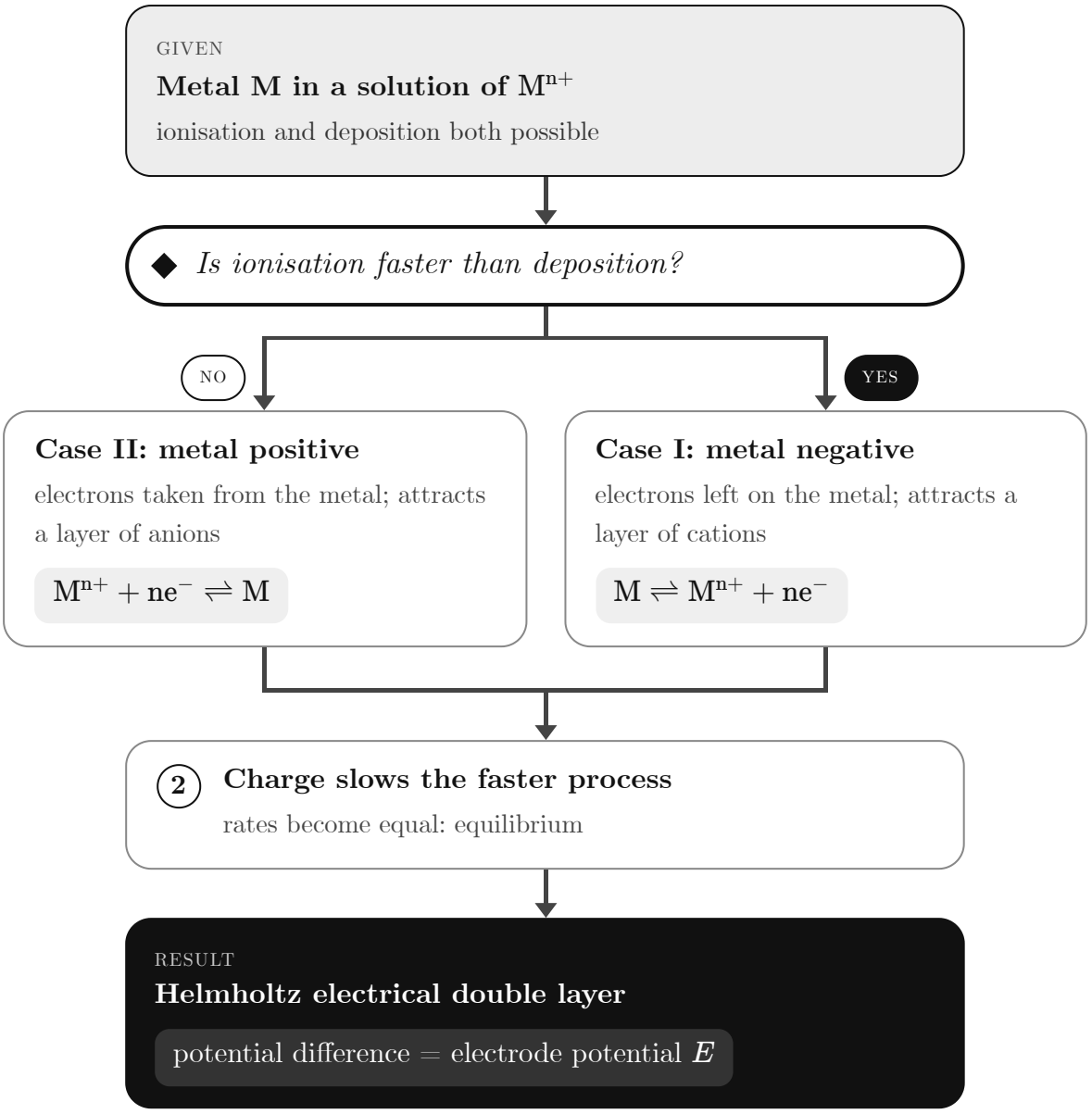


Figure G2.2. Case II: deposition faster than ionisation. ESA ★★

S001·7 · T001 00:08:40

How to read it. Redrawn from slide 001·7: the mirror image of G2.1. The rod carries + charges and attracts a layer of negative ions; write $\mathbf{M^{n+} + ne^- \rightleftharpoons M}$. The bulk anions $\mathbf{A^-}$ are added for clarity.

Flowcharts



Exam trap. Case I: metal (−), solution layer (+). Do not swap them.

Figure F2.1. How the Helmholtz double layer forms ESA ★★★

S001·5 · S001·6 · S001·7 · ESA Jan-25 s2 2b

Case I versus case II of the double layer

	Case I	Case II
Faster reaction	ionisation (oxidation) $M \rightarrow M^{n+} + ne^-$	deposition (reduction) $M^{n+} + ne^- \rightarrow M$
Charge on metal	net negative	net positive
Layer attracted	positive ions	negative ions
Effect of the charge	retards ionisation, increases deposition	retards deposition, increases ionisation
Result	equilibrium + Helmholtz double layer	equilibrium + Helmholtz double layer
Typical metals (added for clarity)	active metals, e.g. Zn	less active metals, e.g. Cu

S001·6 · S001·7

Important points and why

1. Case I: the electrons left behind make the metal negative; it attracts a layer of positive ions (M^{n+}). Case II: the electrons taken from the metal make it positive; it attracts a layer of negative ions. ESA ★★★ S001·6 · S001·7 · ESA Jan-25 s2 2b
2. The charge on the metal is self-limiting: in case I the negative charge retards ionisation and speeds deposition until the two rates are equal (equilibrium); case II is the mirror image. ESA ★★ S001·6 · S001·7
3. Which reaction is faster depends on the nature of the metal (its tendency to oxidise or its ions to be reduced), the ion concentration, the temperature and, for gases, the pressure. ESA ★★ T001 00:10:30

Why does the double layer not keep growing?
In case I the negative charge on the metal pulls M^{n+} back (more deposition) and holds the cations in (less ionisation); the rates become equal and a dynamic equilibrium with a fixed potential difference is reached.
S001·6

Exam trap. In case I the metal is **negative** and the attracted layer is **positive** ions; in case II the metal is positive and the layer is negative ions. Swapping them loses the marks for ESA Jan-25 s2 2.b(i). S001·6 · S001·7 · ESA Jan-25 s2 2b

Worked examples

W2.1 · Double layer when ionisation is faster (3-mark answer) ESA ★★

given Metal M in contact with a solution of M^{n+}; ionisation faster than deposition. Explain the formation of the electrical double layer.

step 1 Two reactions are possible: ionisation $M \rightleftharpoons M^{n+} + ne^-$ and deposition $M^{n+} + ne^- \rightleftharpoons M$.

step 2 Ionisation is faster: M^{n+} ions pass into solution and the electrons stay on the metal, so the metal acquires a net negative charge.

step 3 The negative charge retards ionisation and increases deposition until the rates are equal: equilibrium.

step 4 The negatively charged metal attracts a layer of positive ions (M^{n+}) at the interface.

step 5 The two layers of opposite charge form the Helmholtz electrical double layer, which produces the electrode potential. Draw figure G2.1.

answer Metal (−) + layer of M^{n+} (+) = Helmholtz electrical double layer; its potential difference is the electrode potential.

check Matches slide 001·6 (case I) and its diagram.

ESA Jan-25 s2 2b · S001·5 · S001·6

Past-paper questions

ESA Jan 2025 · 3 marks · UE24CY151A set 2, Q2.b(i) · solved in W2.1

When a metal M is in contact with solution containing its ions M^{n+} and if ionization is faster than deposition, explain the formation of the electrical double layer at the interface of metal and solution.

short-note recap

Electrode potential E: the potential at the interface of a metal and a solution of its own ions. Origin: ionisation $M \rightleftharpoons M^{n+} + ne^-$ and deposition $M^{n+} + ne^- \rightleftharpoons M$ compete. Case I (ionisation faster): metal negative, attracts cations. Case II (deposition faster): metal positive, attracts anions. The two charge layers form the Helmholtz electrical double layer.

3 Standard electrode potential, electrochemical cell and cell notation

Electrode potential depends on the metal, the ion concentration, the temperature and (for gas electrodes) the pressure, so electrodes are compared through the standard electrode potential E°: unit concentration, 298 K and 1 atm. A single electrode potential cannot be measured, so two electrodes are coupled into an electrochemical cell. The cell is written in a standard notation, anode first: $\mathrm{Zn(s)\,|\,Zn^{2+}(1\,M)\,||\,Cu^{2+}(1\,M)\,|\,Cu(s)}$ for the Daniell cell.

Key terms and definitions

Standard electrode potential (E°). The potential developed at the interface of metal and solution, when the metal is in contact with a solution of its own ions having unit concentration at 298 K. In case of gas electrodes the partial pressure of gas is maintained at 1 atmospheric pressure. It is represented as E°. S001·8 · T001 00:11:18

Electrochemical cell. Single electrode potentials cannot be measured, hence two electrodes are coupled together to form a cell. S001·8 · T001 00:12:06

Cell notation. A short representation of a cell: anode on the left, cathode on the right; a single line | separates two phases and a double line || is the salt bridge between the two solutions. S001·8 · T001 00:12:25

Formulas

E3.1 · Standard conditions for E° ESA ★★

$$E^\circ = E \ \text{ at } \ [\mathrm{M^{n+}}] = 1\,\mathrm{M}, \quad T = 298\,\mathrm{K}, \quad p_{gas} = 1\,\mathrm{atm}$$

gives	The standard electrode potential, which depends only on the nature of the electrode.
applies	Comparing electrodes; tabulated in the electrochemical series.
units	V
note	Tabulated as standard reduction potentials.

S001·8 · T001 00:11:27

E3.2 · Cell notation (Daniell cell) ESA ★★★

$$\underbrace{\mathrm{Zn(s)\,|\,Zn^{2+}(1\,M)}}_{\text{anode (LHS)}}\ ||\ \underbrace{\mathrm{Cu^{2+}(1\,M)\,|\,Cu(s)}}_{\text{cathode (RHS)}}$$

gives	A compact representation of the cell.
applies	Writing any cell: anode first, salt bridge in the middle.
units	—
note	The slide writes / and // for | and ||; both are accepted.

S001·8 · T001 00:12:38

Graphs and diagrams

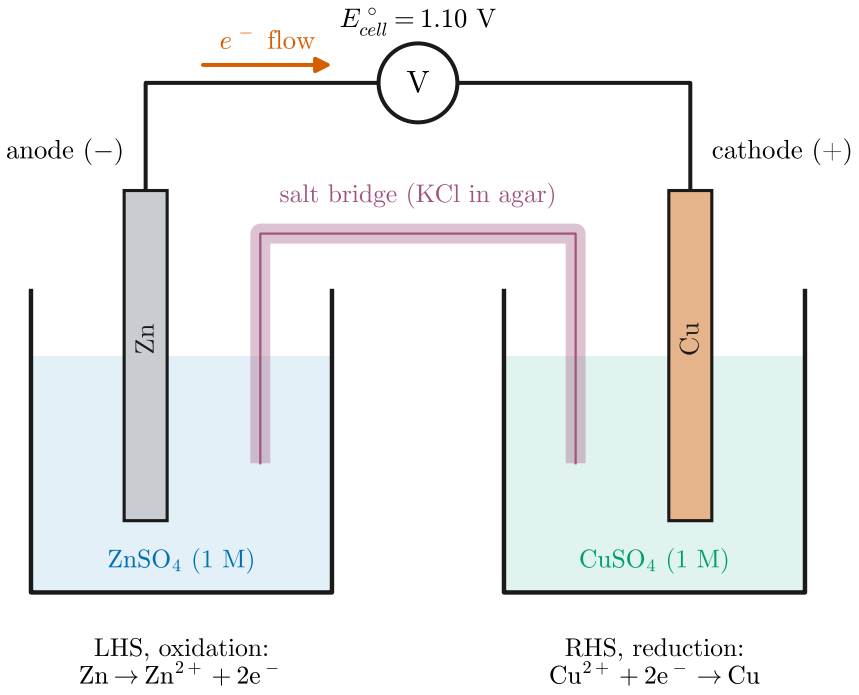


Figure G3.1. The Daniell cell. ESA ★★

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How to read it. Zinc in $\mathbf{ZnSO_4}$ (anode, −, left) and copper in $\mathbf{CuSO_4}$ (cathode, +, right), joined by a salt bridge between the solutions. Electrons flow in the external wire from Zn to Cu; the voltmeter reads $E^\circ_{cell} = 1.10$ V at 1 M. The slide gives only the notation; the diagram is added. Schematic, not to scale.

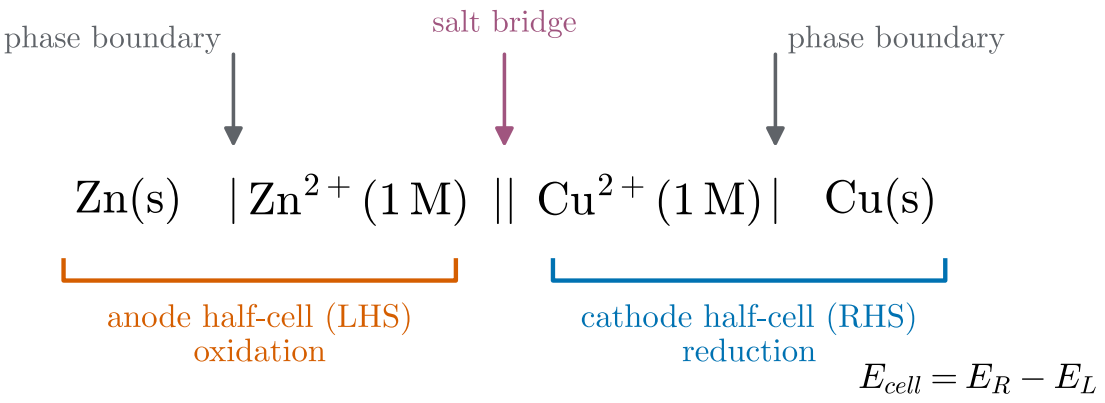


Figure G3.2. Reading the cell notation. ESA ★★★

S001·8 · T001 00:13:13

How to read it. Left half = anode (oxidation), right half = cathode (reduction). Each | is a phase boundary (solid metal | solution); || is the salt bridge, placed between the two solutions. E_{cell} = right minus left.

Four potentials compared

Quantity	Symbol	What it is	Conditions
Electrode potential	E	potential at one metal–solution interface	any; not measurable alone
Standard electrode potential	E°	same, under standard conditions	1 M, 298 K, 1 atm (gases)
Cell potential (emf)	E_{cell}	difference of the two electrode potentials	any concentrations
Standard cell potential	E°_{cell}	$E^\circ_{cathode} - E^\circ_{anode}$	1 M, 1 atm, 298 K

S001·4 · S001·8 · S001·9

Important points and why

1. E° depends only on the nature of the metal, because concentration, temperature and pressure are fixed. ESA ★★ T001 00:11:56
2. In the notation the anode is written first (left), the cathode last (right). In the Daniell cell zinc is oxidised, so Zn is the anode. ESA ★★★ S001·8 · T001 00:12:55
3. Do not write Zn | Zn^{2+} || Cu | Cu^{2+}: the salt bridge connects the two *solutions*, so the ions sit next to ||. ESA ★★ T001 00:13:43
4. The concentration of each ion is written in brackets after it, e.g. $Zn^{2+}(1\,M)$. ESA ★★ S001·8 · T001 00:13:13

Why can a single electrode potential not be measured?
A voltmeter measures a potential *difference*; connecting it to the solution needs a second conductor in the solution, which is itself a second electrode. So two electrodes are always coupled into a cell, and E° values are measured against a reference (SHE = 0).

ADDED FOR CLARITY S001·8 · T001 00:12:06

Exam trap. The anode goes on the **left**. Writing the Daniell cell as Cu | $\mathrm{Cu^{2+}}$ || $\mathrm{Zn^{2+}}$ | Zn gives $E_{cell} = -1.10$ V, a non-spontaneous cell. ADDED FOR CLARITY S001·8 · S001·9

Worked examples

W3.1 · Cell notation and half-reactions of the Daniell cell ESA ★★★

GIVEN	Zinc dipped in 1 M $\mathrm{ZnSO_4}$ and copper dipped in 1 M $\mathrm{CuSO_4}$, joined by a salt bridge.
STEP 1	From the series, Zn (-0.76 V) is lower than Cu ($+0.34$ V), so Zn is the anode (oxidation) and Cu the cathode (reduction).
STEP 2	Anode: $\mathrm{Zn \rightarrow Zn^{2+} + 2e^-}$. Cathode: $\mathrm{Cu^{2+} + 2e^- \rightarrow Cu}$.
STEP 3	Overall: $\mathrm{Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu}$, $n = 2$.
STEP 4	Anode left, | for each phase boundary, || for the salt bridge.
ANSWER	$\boxed{\mathrm{Zn(s)\,|\,Zn^{2+}(1\,M)\,||\,Cu^{2+}(1\,M)\,|\,Cu(s)}}$
CHECK	Same as slide 001·8 (which writes / and //).

S001·8 · T001 00:12:38

SHORT-NOTE RECAP

E°: electrode potential at unit concentration, 298 K and 1 atm for gases; depends only on the electrode. Two electrodes form a cell (single potentials are not measurable). Notation: anode | anode solution || cathode solution | cathode, e.g. $\mathrm{Zn|Zn^{2+}||Cu^{2+}|Cu}$.

4 Cell potential, standard cell potential and free energy

The cell potential E_{cell} is the difference between the electrode potentials of the two electrodes; under standard conditions (1 M, 1 atm, 298 K) it is the standard cell potential E°_{cell}. It is calculated as $E_{cell} = E_{rhs} - E_{lhs} = E_{cathode} - E_{anode}$, using reduction potentials for both. E_{cell} is the driving force of the cell reaction: $\Delta G = -nFE_{cell}$, so a spontaneous reaction ($\Delta G < 0$) has a positive E_{cell}. This convention is used in every cell numerical in the ESA papers.

Key terms and definitions

Cell potential (E_{cell}). The difference in electrode potentials of the electrodes constituting the cell. It is denoted by E_{cell}. S001·9 · T001 00:14:25

Standard cell potential (E°_{cell}). When all the concentrations are 1 M, all partial pressures of gases are 1 atm and temperature is 298 K, the emf is called the standard cell potential, E°_{cell}. S001·9 · T001 00:15:01

> *"$E_{cell} = E_{rhs} - E_{lhs} = E_{cathode} - E_{anode}$. E_{cell} represents the driving force for the cell reaction to take place: $\Delta G = -nFE_{cell}$. If the reaction is spontaneous ΔG is negative, thus E_{cell} should be positive; if the reaction is non-spontaneous ΔG is positive, thus E_{cell} should be negative."* S001·9

Formulas

E4.1 · Cell potential ESA ★★★

$$E_{cell} = E_{rhs} - E_{lhs} = E_{cathode} - E_{anode}$$

gives	The emf of the cell from the two electrode (reduction) potentials.
applies	Every cell; with E° values it gives E°_{cell}.
units	V
note	Use reduction potentials for both electrodes; do not change the sign of the anode value before subtracting.

S001·9 · T001 00:15:37

E4.2 · Standard cell potential ESA ★★★

$$E^{\circ}_{cell} = E^{\circ}_{cathode} - E^{\circ}_{anode}$$

gives	The emf when all concentrations are 1 M, gases at 1 atm, $T = 298$ K.
applies	First step of every cell numerical (then Nernst, class 002).
units	V
note	Daniell: $0.34 - (-0.76) = 1.10$ V.

S001·9 · T001 00:15:20

E4.3 · Free energy and cell potential ESA ★★★

$$\Delta G = -nFE_{cell} \qquad \Delta G^\circ = -nFE^\circ_{cell}$$

GIVES The free-energy change of the cell reaction (= minus the maximum electrical work).

APPLIES Deciding spontaneity; converting emf to energy.

UNITS ΔG in J mol^{-1} (E in V, F in C mol^{-1})

NOTE n = moles of electrons transferred in the balanced cell reaction; $F = 96500$ C mol^{-1}. The standard form $\Delta G^\circ = -nFE^\circ$ is stated in the class 002 derivation.

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E4.4 · Spontaneity criterion ESA ★★★

$$E_{cell} > 0 \;\Leftrightarrow\; \Delta G < 0 \text{ (spontaneous)} \qquad E_{cell} < 0 \;\Leftrightarrow\; \Delta G > 0 \text{ (non-spontaneous)}$$

GIVES Whether the cell reaction as written proceeds on its own.

APPLIES After computing E_{cell}.

UNITS —

NOTE Galvanic cells run on reactions with $E_{cell} > 0$; electrolytic cells force reactions with $E_{cell} < 0$.

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Graphs and diagrams

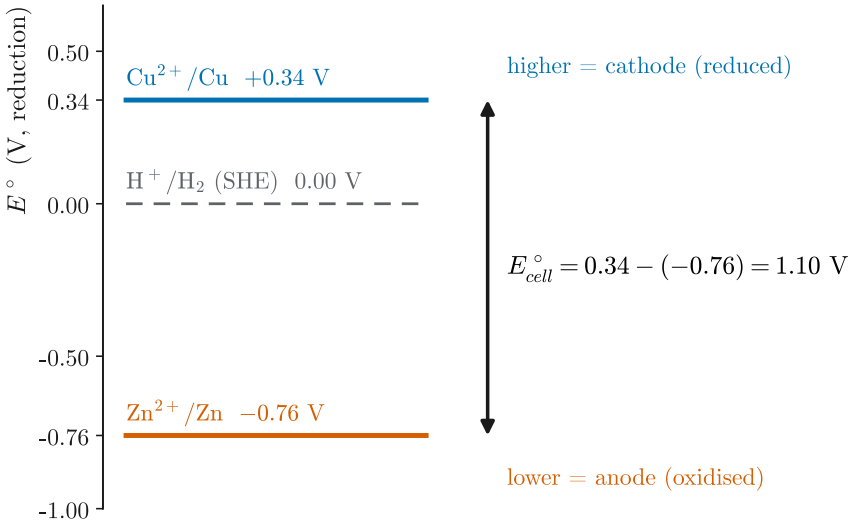
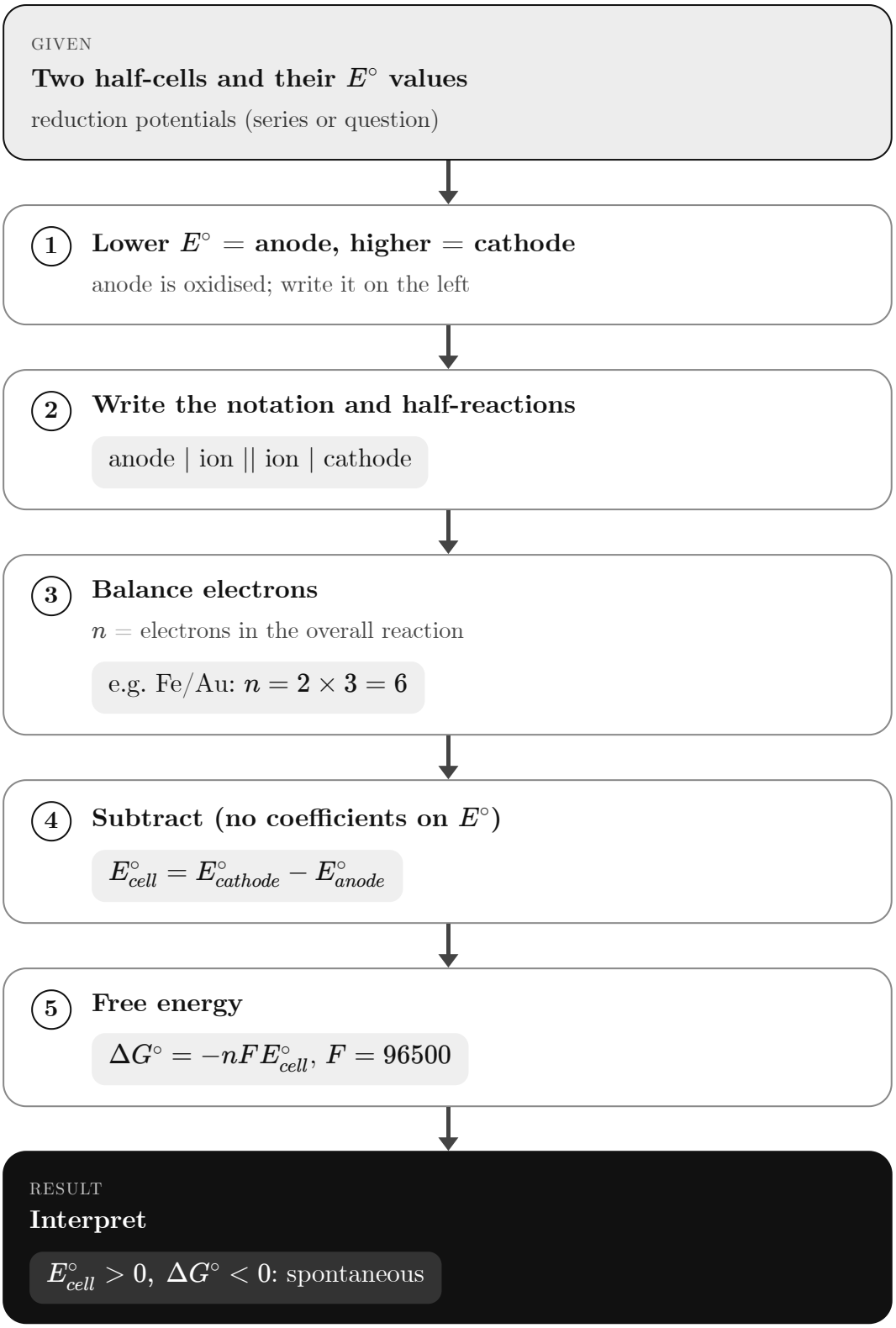


Figure G4.1. E°_{cell} as the gap between two electrodes on the potential scale. ESA ★★★

ADDED FOR CLARITY S001·9 · S001·10

How to read it. Place both reduction potentials on a vertical scale. The upper one is the cathode, the lower the anode, and E°_{cell} is the (positive) distance between them: $0.34 - (-0.76) = 1.10$ V for the Daniell cell.

Flowcharts



Remark. For non-standard concentrations continue with the Nernst equation (class 002); the ESA cell numericals (session 006) all start with steps 1–4.

Figure F4.1. From two electrodes to E°_{cell} and ΔG° ESA ★★★

S001·8 · S001·9 · S001·10

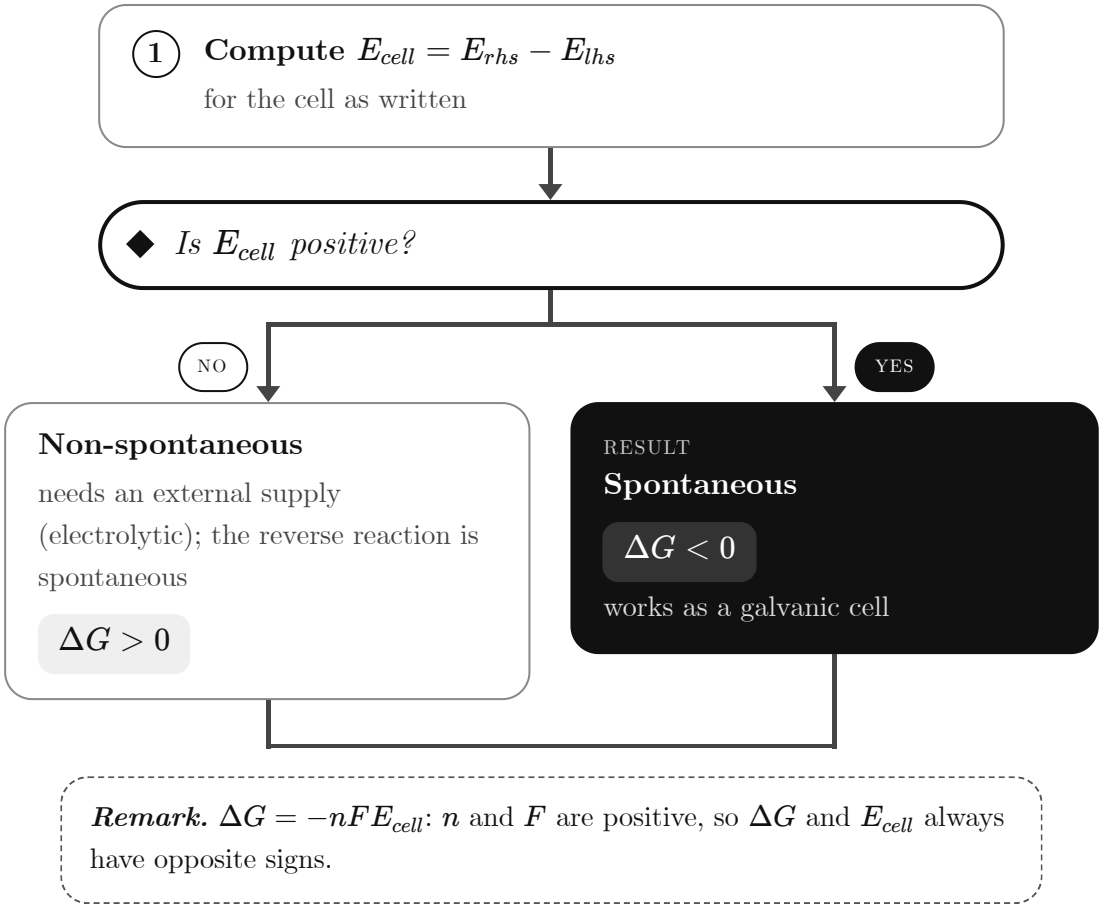


Figure F4.2. Spontaneous or not? ESA ★★★

S001·9 · T001 00:16:39

Important points and why

1. $E_{cell} = E_{cathode} - E_{anode}$ (right minus left) is used in every cell numerical of the UE23/UE24 papers (Q2), so this convention is examined every time. ESA ★★★ S001·9 · ESA Jul-23a 2c · ESA Dec-23 2c · ESA Jun-24b 2a
2. E_{cell} depends on the concentrations, the temperature and the partial pressures of gases; E°_{cell} fixes them at 1 M, 1 atm and 298 K. ESA ★★ S001·9 · T001 00:14:43
3. E_{cell} is the driving force of the cell reaction: a system moves toward minimum free energy, and $\Delta G = -nFE_{cell}$. ESA ★★★ S001·9 · T001 00:16:15
4. E°_{cell} is intensive: it does not change when the cell reaction is multiplied by a number, but ΔG° does (through n). ESA ★★ ADDED FOR CLARITY S001·9

Why is $\Delta G = -nFE$?
The maximum work a cell can do equals the total charge times the energy per unit charge: $W_{max} = nF \times E$; the maximum (non-expansion) work equals $-\Delta G$. Derived in class 002.
T001 00:20:11 · T001 00:21:06

Why subtract reduction potentials instead of adding an oxidation potential?
Both are equivalent: $E^\circ_{ox}(\text{anode}) = -E^\circ_{red}(\text{anode})$, so $E^\circ_{red,cathode} + E^\circ_{ox,anode} = E^\circ_{cathode} - E^\circ_{anode}$. The slides tabulate reduction potentials, so subtract.

ADDED FOR CLARITY S001·9 · S001·10

Exam trap. Never double-flip signs: in $E^\circ_{cathode} - E^\circ_{anode}$ use the tabulated reduction potentials as they are. $0.34 - (-0.76) = 1.10$ V, not $0.34 - 0.76$. ADDED FOR CLARITY S001·9

Exam trap. Do not multiply E° by stoichiometric coefficients when balancing (e.g. $2\text{Al} + 3\text{Cu}^{2+}$); only n in $\Delta G = -nFE$ changes. ADDED FOR CLARITY ESA Jan-25 s1 2a

Exam trap. $\Delta G = -nFE$ gives joules per mole of reaction; divide by 1000 for kJ. n is the number of electrons in the balanced overall reaction (6 for Fe/Au, 2 for Daniell). ADDED FOR CLARITY S001·9

Worked examples

W4.1 · E°cell and ΔG° of the Daniell cell — ESA ★★★

GIVEN $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76$ V, $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34$ V, $F = 96500$ C mol^{-1}.

STEP 1 $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = 0.34 - (-0.76) = 1.10$ V.

STEP 2 $n = 2$ (two electrons per Zn).

STEP 3 $\Delta G^\circ = -nFE^\circ_{cell} = -2 \times 96500 \times 1.10 = -212\,300$ J mol^{-1}.

ANSWER $\boxed{E^\circ_{cell} = 1.10 \text{ V};\ \Delta G^\circ = -212.3 \text{ kJ mol}^{-1} \text{ (spontaneous)}}$

CHECK sympy: 1.10 V and −212 300 J exactly.

S001·9 · S001·10

W4.2 · The same cell written the wrong way round — ESA ★★

GIVEN $\text{Cu(s)} \,|\, \text{Cu}^{2+}(1\,\text{M}) \,||\, \text{Zn}^{2+}(1\,\text{M}) \,|\, \text{Zn(s)}$.

STEP 1 $E^\circ_{cell} = E^\circ_{rhs} - E^\circ_{lhs} = -0.76 - 0.34 = -1.10$ V.

STEP 2 $\Delta G^\circ = -2 \times 96500 \times (-1.10) = +212\,300$ J mol^{-1}.

ANSWER $\boxed{E^\circ_{cell} = -1.10 \text{ V},\ \Delta G^\circ = +212.3 \text{ kJ mol}^{-1}\text{: non-spontaneous as written}}$

CHECK Reverse of W4.1: same magnitude, opposite sign; it would need an external supply (electrolytic).

ADDED FOR CLARITY S001·9

W4.3 · Silver–copper cell: E°cell and ΔG° ESA ★★★

GIVEN A silver electrode in $\mathrm{AgNO_3}$ and a copper electrode in $\mathrm{CuSO_4}$ joined by a salt bridge; $E^\circ(\mathrm{Ag^+/Ag}) = 0.80$ V, $E^\circ(\mathrm{Cu^{2+}/Cu}) = 0.34$ V.

STEP 1 Cu has the lower E°: anode. Ag: cathode.

STEP 2 Cell: $\mathrm{Cu\,|\,Cu^{2+}\,||\,Ag^+\,|\,Ag}$; overall $\mathrm{Cu + 2Ag^+ \rightarrow Cu^{2+} + 2Ag}$, $n = 2$.

STEP 3 $E^\circ_{cell} = 0.80 - 0.34 = 0.46$ V.

STEP 4 $\Delta G^\circ = -2 \times 96500 \times 0.46 = -88\,780$ J mol^{-1}.

ANSWER $\boxed{E^\circ_{cell} = 0.46 \text{ V};\ \Delta G^\circ = -88.78 \text{ kJ mol}^{-1}}$

CHECK sympy. The E_{cell} part of the ESA question (0.15 M, 0.19 M) needs the Nernst equation: classes 002 and 006.

ESA Jul-23a 2c · S001·10

W4.4 · Iron–gold cell: slide value 1.50 V versus ESA value 1.52 V ESA ★★★

GIVEN $\mathrm{Fe\,|\,Fe^{2+}\,||\,Au^{3+}\,|\,Au}$; $E^\circ(\mathrm{Fe^{2+}/Fe}) = -0.44$ V; $E^\circ(\mathrm{Au^{3+}/Au}) = 1.52$ V (papers) or 1.50 V (slide).

STEP 1 Anode $\mathrm{Fe \rightarrow Fe^{2+} + 2e^-}$ (×3); cathode $\mathrm{Au^{3+} + 3e^- \rightarrow Au}$ (×2):
$\mathrm{3Fe + 2Au^{3+} \rightarrow 3Fe^{2+} + 2Au}$, $n = 6$.

STEP 2 With 1.52 V: $E^\circ_{cell} = 1.52 - (-0.44) = 1.96$ V; $\Delta G^\circ = -6 \times 96500 \times 1.96 = -1\,134\,840$ J mol^{-1}.

STEP 3 With 1.50 V: $E^\circ_{cell} = 1.94$ V; $\Delta G^\circ = -1\,123\,260$ J mol^{-1}.

ANSWER Paper value: $E^\circ_{cell} = 1.96$ V, $\Delta G^\circ = -1134.8$ kJ mol^{-1} (slide value: 1.94 V, -1123.3 kJ mol^{-1})

CHECK sympy. Use the E° printed in the question; 1.96 V also matches the consolidated slides (problem 7).

ESA Dec-23 2c · ESA Jun-24b 2a · S001·10

W4.5 · Aluminium–copper cell from the series ESA ★★

GIVEN $2Al + 3Cu^{2+} \rightarrow 2Al^{3+} + 3Cu$; series values -1.66 V and $+0.34$ V.

STEP 1 Al is lower: anode. $E^\circ_{cell} = 0.34 - (-1.66) = 2.00$ V (no multiplying by 2 or 3).

STEP 2 $n = 6$ (2 Al × 3 e^-).

STEP 3 $\Delta G^\circ = -6 \times 96500 \times 2.00 = -1\,158\,000$ J mol^{-1}.

ANSWER $E^\circ_{cell} = 2.00$ V; $\Delta G^\circ = -1158$ kJ mol^{-1}

CHECK The ESA question gives $E^\circ_{cell} = 2.0$ V, which agrees with the series. Its Nernst part is in class 002.

ESA Jan-25 s1 2a · S001·10

Past-paper questions

ESA Jul 2023 · 5 marks · UE22CY151B Q2.c (E°cell part) · solved in W4.3

A silver electrode containing 0.15 M silver nitrate and copper electrode containing 0.19 M copper sulphate were coupled using salt bridge. Write the cell representation, half cell reactions, calculate E°_{cell}, E_{cell} at 25 °C. (Primary session 006; the E°_{cell} step uses this class.)

ESA Dec 2023 / Jun 2024 · 5 marks · UE22CY151B Q2.c; UE23CY151B Q2.a (E°cell part) · solved in W4.4

$Fe/Fe^{2+}(0.1\,M)\,||\,Au^{3+}(0.5\,M)/Au$: half-cell reactions, E°_{cell} and E_{cell} at 298 K ($E^\circ_{Au^{3+}/Au} = 1.52$ V, $E^\circ_{Fe^{2+}/Fe} = -0.44$ V). Dec-23: 5 marks; Jun-24b: 6 marks. (Primary session 006.)

SHORT-NOTE RECAP

E_{cell} = difference of the electrode potentials = $E_{rhs} - E_{lhs} = E_{cathode} - E_{anode}$; standard (1 M, 1 atm, 298 K): E°_{cell}. $\Delta G = -nFE_{cell}$: spontaneous $\Leftrightarrow$ $\Delta G < 0$ $\Leftrightarrow$ $E_{cell} > 0$. Daniell: $E^\circ_{cell} = 1.10$ V, $\Delta G^\circ = -212.3$ kJ mol^{-1}.

5 The electrochemical series and its uses

Arranging electrodes in order of their standard reduction potentials gives the electrochemical series, from lithium (-3.03 V) to gold ($+1.50$ V on the slide). A negative value indicates an oxidation tendency and a positive value a reduction tendency. The series predicts which electrode is the anode, whether a reaction is spontaneous, and which metal corrodes when two metals are in contact (galvanised versus tinned iron).

Key terms and definitions

Electrochemical series. In order to predict the electrochemical behaviour of an electrode–electrolyte system, elements are arranged in the order of their standard reduction potentials. This arrangement is known as the electrochemical series. S001·10 · T001 00:17:34

> *"A negative value indicates oxidation tendency while a positive value indicates a reduction tendency."* S001·10

Graphs and diagrams

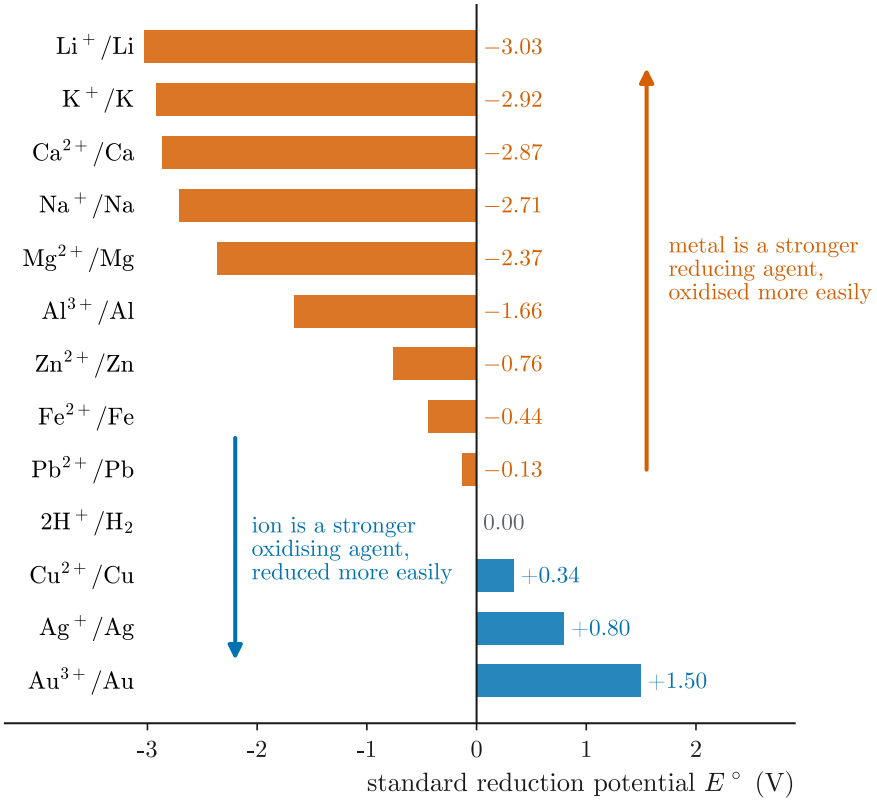
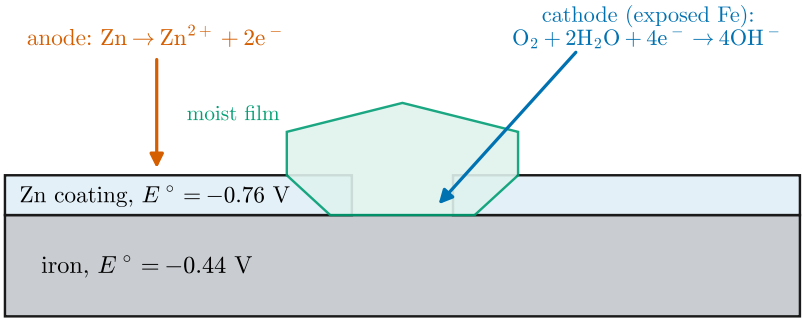


Figure G5.1. The electrochemical series (slide values). ESA ★★

S001·10

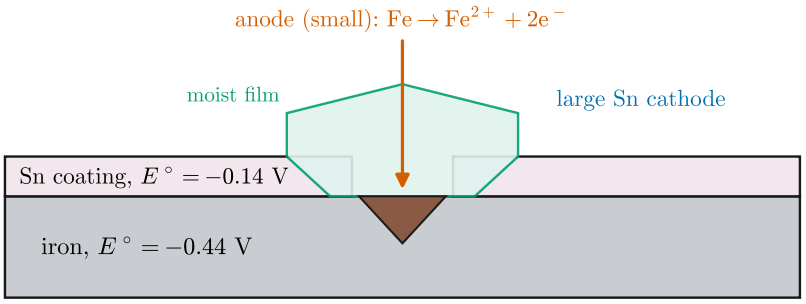
How to read it. Redrawn from the table on slide 001·10, with the same 13 couples and values. Bars to the left (negative E°) mark an oxidation tendency, bars to the right a reduction tendency. Upwards: the metal is the stronger reducing agent; downwards: the ion is the stronger oxidising agent. The slide lists Au at +1.50 V; ESA problems use 1.52 V.

(a) galvanised: Zn more active ⇒ Zn is the anode



iron is protected (sacrificial / cathodic protection) until the zinc is used up

(b) tinned: Fe more active ⇒ exposed Fe is the anode



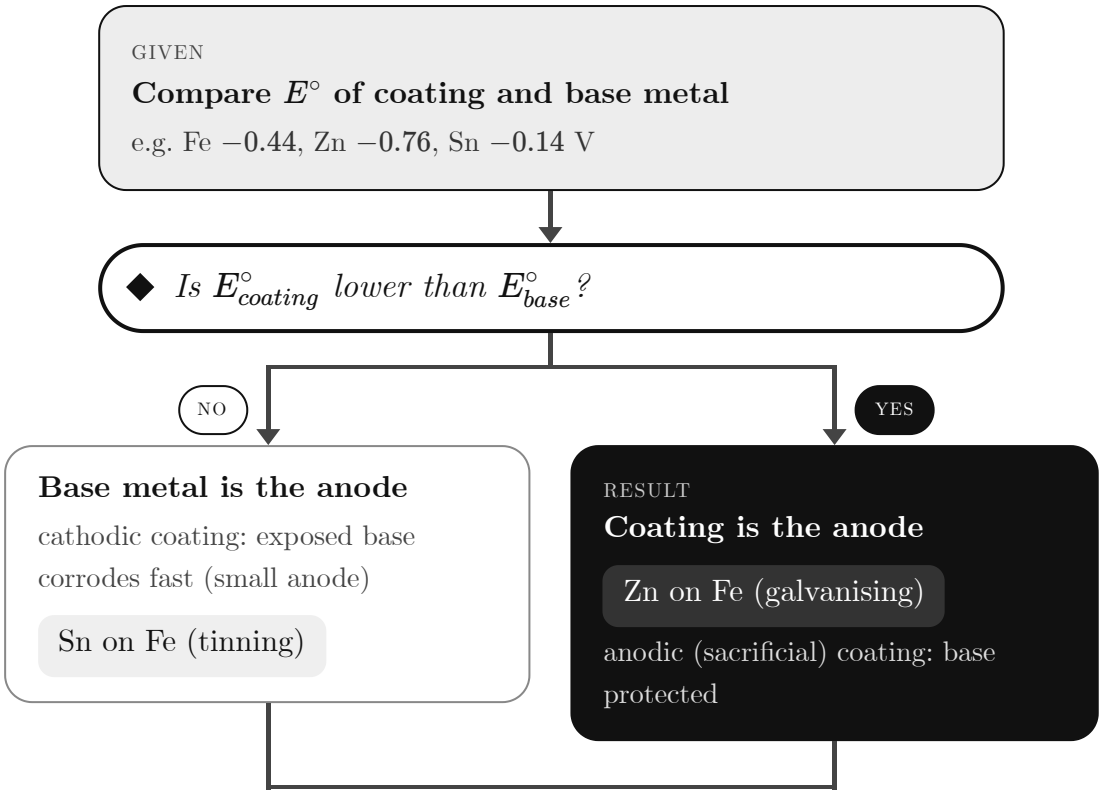
small anode, large cathode: intense local corrosion (pitting) of the exposed iron

Figure G5.2. Scratched zinc and tin coatings on iron. ESA ★★

ADDED FOR CLARITY ESA Jul-23a 4d

How to read it. (a) Zn (**−0.76** V) is below Fe (**−0.44** V) in E°, so zinc is the anode and dissolves while exposed iron is a cathode: iron is protected. (b) Sn (**−0.14** V) is above Fe, so exposed iron is a small anode next to a large tin cathode and pits rapidly. Schematic, not to scale.

Flowcharts



Remark. Coatings are taught in class 010; this chart only applies the electrochemical series.

Figure F5.1. Does the base metal corrode when the coating is scratched? ESA ★★

ESA Jul-23a 4d · S001·10

Electrochemical series (slide 001·10)

Couple (oxidant ⇌ reductant)	E° (V)	Tendency
$Li^+ + e^- \rightleftharpoons Li$	−3.03	strongest reducing metal
$K^+ + e^- \rightleftharpoons K$	−2.92	oxidation
$Ca^{2+} + 2e^- \rightleftharpoons Ca$	−2.87	oxidation
$Na^+ + e^- \rightleftharpoons Na$	−2.71	oxidation
$Mg^{2+} + 2e^- \rightleftharpoons Mg$	−2.37	oxidation
$Al^{3+} + 3e^- \rightleftharpoons Al$	−1.66	oxidation
$Zn^{2+} + 2e^- \rightleftharpoons Zn$	−0.76	oxidation
$Fe^{2+} + 2e^- \rightleftharpoons Fe$	−0.44	oxidation

Couple (oxidant ⇌ reductant)	E° (V)	Tendency
$Pb^{2+} + 2e^- \rightleftharpoons Pb$	-0.13	oxidation
$2H^+ + 2e^- \rightleftharpoons H_2$	0.00	reference (SHE)
$Cu^{2+} + 2e^- \rightleftharpoons Cu$	$+0.34$	reduction
$Ag^+ + e^- \rightleftharpoons Ag$	$+0.80$	reduction
$Au^{3+} + 3e^- \rightleftharpoons Au$	$+1.50$ (ESA: 1.52)	strongest oxidising ion

S001·10

Zinc coating versus tin coating on iron

	Galvanising (Zn)	Tinning (Sn)
E° of coating vs Fe (-0.44 V)	-0.76 V: lower	-0.14 V: higher
At a scratch, the anode is	zinc	iron
Type of coating	anodic (sacrificial)	cathodic
Iron at the scratch	protected	corrodes rapidly (pitting)

ADDED FOR CLARITY ESA Jul-23a 4d · S001·10

Important points and why

1. The series runs from Li (-3.03 V, oxidised most easily) to Au ($+1.50$ V on the slide, reduced most easily). Hydrogen (0.00 V) is the reference. ESA ★★ S001·10 · T001 00:18:21
2. Of two electrodes, the one with the lower (more negative) E° is the anode, and E°_{cell} is the gap between them on the series. ESA ★★★ ADDED FOR CLARITY S001·10
3. A metal above hydrogen (negative E°, e.g. Zn) liberates H_2 from dilute acid; one below hydrogen (Cu, Ag, Au) does not. ESA ★★ ADDED FOR CLARITY S001·10
4. A metal displaces from solution the ions of any metal below it in the series (Zn + Cu^{2+} → Zn^{2+} + Cu). ESA ★★ ADDED FOR CLARITY S001·10
5. Corrosion: when two metals touch in a moist environment, the one with the lower E° becomes the anode and corrodes (used again in the corrosion classes 006–011). ESA ★★ ADDED FOR CLARITY ESA Jul-23a 4d

Why does a negative E° mean an oxidation tendency?
Coupled with the hydrogen electrode, such a metal gives $E^\circ_{cell} > 0$ only as the anode, i.e. when it is oxidised; Li (-3.03 V) oxidises very easily while Cu^{2+} ($+0.34$ V) is easily reduced.
S001·10 · T001 00:18:05

Exam trap. The slide prints $E^\circ(\mathrm{Au^{3+}/Au}) = +1.50$ V, but the ESA Fe/Au problems give 1.52 V. Always use the value printed in the question. S001·10 · ESA Dec-23 2c · ESA Jun-24b 2a

Exam trap. The slide's right-hand arrow reads "Metal Oxidizing Activity Increasing" downwards; it is the *ion* (oxidant) whose oxidising power increases downwards. The metal's tendency to be oxidised increases *upwards*. ADDED FOR CLARITY S001·10

Exam trap. A tin coating protects iron only while it is intact: Sn (-0.14 V) is above Fe (-0.44 V) in E°, so at a scratch iron becomes the anode and corrodes faster. ADDED FOR CLARITY ESA Jul-23a 4d

Worked examples

W5.1 · Which metal liberates hydrogen from 1 M acid? ESA ★★

GIVEN Zn and Cu in 1 M $\mathrm{H^+}$; E°: Zn -0.76, $\mathrm{H^+/H_2}$ 0.00, Cu $+0.34$ V.

STEP 1 Zn + $\mathrm{2H^+}$: Zn anode, $\mathrm{H_2}$ cathode: $E^\circ_{cell} = 0 - (-0.76) = +0.76$ V; $\Delta G^\circ = -146.68$ kJ mol^{-1}.

STEP 2 Cu + $\mathrm{2H^+}$: $E^\circ_{cell} = 0 - 0.34 = -0.34$ V; $\Delta G^\circ = +65.62$ kJ mol^{-1}.

ANSWER Zn liberates $\mathrm{H_2}$ (spontaneous); Cu does not

CHECK sympy; agrees with the rule that metals above hydrogen displace it.

ADDED FOR CLARITY S001·10

W5.2 · Galvanised versus tinned iron when the coating wears off ESA ★★

GIVEN Reduction potentials: Fe -0.44 V, Zn -0.76 V, Sn -0.14 V. Will iron corrode if a section of the coating wears off?

STEP 1 Galvanised (Zn on Fe): Zn has the lower E°, so Zn is the anode and Fe the cathode; couple $E^\circ = -0.44 - (-0.76) = 0.32$ V.

STEP 2 Zn corrodes sacrificially ($\mathrm{Zn \rightarrow Zn^{2+} + 2e^-}$); exposed iron only carries the cathodic reaction ($\mathrm{O_2 + 2H_2O + 4e^- \rightarrow 4OH^-}$). No corrosion of iron while Zn remains.

STEP 3 Tinned (Sn on Fe): Fe has the lower E°, so the exposed Fe is the anode and Sn the cathode; couple $E^\circ = -0.14 - (-0.44) = 0.30$ V.

STEP 4 Small anode (scratch) and large cathode (tin surface): the anodic current density is high, so iron corrodes rapidly at the scratch (pitting).

ANSWER Galvanised: iron not corroded (zinc is sacrificed). Tinned: iron corrodes, faster than if it were uncoated at that spot.

CHECK sympy: 0.32 V and 0.30 V, both positive, so in each couple the lower-E° metal is oxidised. Full treatment: class 010 (metal coatings).

ESA Jul-23a 4d · S001·10

Past-paper questions

ESA Jul 2023 · 2 marks · UE22CY151B Q4.d(i) · solved in W5.2

Galvanization and tinning are two popular methods of protecting a metal by coating with another metal. If the reduction potentials of iron, zinc and tin are −0.44 V, −0.76 V and −0.14 V respectively, then comment on the situation: corrosion of iron would be observed in case a section of the protecting layer wears off. Briefly discuss the reason. (4.d is 4 marks with part ii.)

SHORT-NOTE RECAP

Electrochemical series: electrodes in order of standard reduction potential, Li -3.03 ... $\mathrm{H_2}$ 0.00 ... Au $+1.50$ V (slide). Negative = oxidation tendency (strong reducing metals); positive = reduction tendency. Lower E° = anode. Uses: E°_{cell}, spontaneity, displacement, corrosion (Zn protects Fe; Sn only while intact).

A Symbol glossary

Symbol	Name	Meaning	Unit	Appears in
E	Electrode potential	Potential at a metal–solution interface	V	T2
E°	Standard electrode potential	Standard reduction potential (1 M, 298 K, 1 atm)	V	E3.1, E4.2
E_{cell}	Cell potential (emf)	$E_{cathode} - E_{anode}$	V	E4.1, E4.3
E°_{cell}	Standard cell potential	$E^\circ_{cathode} - E^\circ_{anode}$	V	E4.2
E_{rhs}, E_{lhs}	Right / left electrode	Cathode / anode in the cell notation	V	E4.1
ΔG, ΔG°	Gibbs free-energy change	Of the cell reaction (standard: ΔG°)	J mol^{-1}	E4.3
n	Electrons transferred	Moles of electrons per mole of overall reaction; in $\mathrm{M^{n+}}$ it is the ion charge (the same n for a single electrode)	—	E2.1, E4.3
F	Faraday constant	Charge on one mole of electrons, 96500 (96485)	C mol^{-1}	E4.3
M, $\mathrm{M^{n+}}$	Metal and its ion	Electrode metal and its ions in solution	—	E2.1
| , ||	Phase boundary, salt bridge	Cell notation (slides: / and //)	—	E3.2

B Sources, coverage and gaps

Source	Path	Use
Deck 001 (.pptx rendered at 80 dpi; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.*	11 slides; content on 3–10
Transcript 001 (.transcript.md, .transcript.txt)	pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/001_Origin_of_electrode_potential_cell_potential__3e2916789bf0cc7b.*	25.9 min lecture; 00:00–18:50 used (the rest is Nernst, class 002)
Consolidated Unit 2 slides (older syllabus)	pesuc/First_Year_8enterprise/Files/Chemistry/Slides/Chem_Slides_Unit-2__1C1NjUHA4j.pdf	Same slides 3–10; problem 7 (Fe/Au, 1.52 V); series uses in corrosion
ESA papers (13 files, 10 unique)	pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/*/*.pdf	Jan-25 s2 2.b(i); Jul-23a 4.d(i); E°cell steps of Jul-23a 2.c, Dec-23 2.c, Jun-24b 2.a, Jan-25 s1 2.a
Video 001 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/001_*.mp4	Not opened: every slide was readable in the render; no video trigger applied

Ten unique ESA papers were checked (UE22CY151B Jul 2023 a/b, Dec 2023, May 2024, Dec 2024; UE23CY151A/B Jan 2024 set 2, Jun 2024 a/b, Dec 2024; UE24CY151A Jan 2025 sets 1 and 2; three 2024 files are one paper). Directly on this class: Jan 2025 set 2 Q2.b(i), 3 marks (double layer, case I). The electrochemical series is applied in Jul 2023 Q4.d(i) (zinc versus tin coating; primary class 010). $E_{cell} = E_{cathode} - E_{anode}$ is used in every UE23/UE24 Q2 cell numerical (Jul-23a 2.c, Dec-23 2.c, Jun-24b 2.a and others; primary class 006), so the convention is rated ★★★. The galvanic/electrolytic comparison, cell notation as a stand-alone question, the standard-electrode-potential definition and $\Delta G = -nFE$ as a separate question were never asked.

Known gaps, conflicts and corrections

1. No class-work (CW) sheet for Chemistry exists in the bucket; worked examples are built from the slides and ESA papers and verified with sympy.
2. Transcript 001 also covers class 002 (Nernst equation, from 00:18:50); that part is left to pack chem-u2-002.
3. Slide 001·10 (series table, image only) prints “Silver: A+(aq)” for $\mathbf{Ag^{+}}$; corrected.
4. Slide 001·10 gives $\boldsymbol{E^{\circ}(Au^{3+}/Au) = +1.50}$ V; the ESA Fe/Au problems (Dec-23 2.c, Jun-24b 2.a) and consolidated-slide problem 7 use 1.52 V. Both are shown (W4.4). Li ($\mathbf{-3.03}$) and K ($\mathbf{-2.92}$) are also slightly off the usual tabulated $\mathbf{-3.04}$ and $\mathbf{-2.93}$ V; slide values are kept.
5. Slide 001·10's right-hand arrow says “Metal Oxidizing Activity Increasing” downwards; it is the oxidising power of the ion that increases downwards. Clarified as an exam trap.

6. The slides spell the cell “Daniel cell” and use / and // in the notation; written here as Daniell and | , || (both notations are accepted).
7. The deck has no cell diagram; the Daniell cell, electrolytic cell and potential-scale figures are added for clarity.
8. The slide says single electrode potentials cannot be measured but does not name the reference; SHE = 0.00 V is added for clarity (it is the 0.00 V row of the series).
9. At T001 00:15:01 the lecturer says “standard electrode potential” where the standard cell potential is meant; the slide wording is followed.
10. Transcript recognition errors corrected silently: “detox/redux reaction” → redox reaction; “hellholes / And molds / Helmor's electric double layer” → Helmholtz electrical double layer; “ionside” → ions; “zirconium oxide dope by two or three” → zirconia doped with Y2O3; “Daniel Cell” → Daniell cell; “zipped” → dipped.

Coverage

Source	Citations	Slides cited (slide × citations)
S001	92	3×5, 4×6, 5×4, 6×11, 7×9, 8×14, 9×21, 10×22
ESA	21	
T001	48	

Slides 001·1, 001·2 and 001·11 are title, contents and closing slides. The deck says “Class 14 content” (older numbering); it is session 001 of Unit 2 in the bucket. Transcript T001 covers classes 001 and 002: this pack uses 00:00:00–00:18:50; 00:18:50–00:25:55 (Nernst equation) belongs to pack chem-u2-002, and is cited here only for $W_{max} = nFE$. The series values on slide 001·10 exist only in the slide image.